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Highly Reliable Subsurface Compressor System Enables Increased Natural Gas Production and Recoverability

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Abstract

The primary objective of this paper is to examine the performance of an electrically driven, high-speed downhole compressor system to increase the rate of natural gas production and extend the economic usefulness of natural gas wells relative to traditional surface wellhead compressors. By developing and integrating enabling technology building blocks to the compression system, effective drawdown near the perforations and increased delivery of reserves are feasible and demonstrated in several field trials.

The Subsurface Compression System (SCS) for gas wells is comprised of a novel architecture developed for increasing gas production with superior downhole reliability to reduce the production downtime and workover costs commonly caused by failures of equipment. There are three core technology building blocks forming the novel architecture of the subsurface compressor. A hermetically sealed, high-speed permanent magnet motor block, which completely isolates all electrical components from gasses and liquids present in the external environment and can spin up to 50,000 Revolutions per Minute (RPM). A wide frequency Variable Speed Drive (VSD), located on the surface, drives the downhole motor with wide speed ranges to accommodate changing well conditions throughout the life of the subsurface compressor. A magnetic coupling block that maintains a hermetic seal of the motor while effectively transmitting torque from the motor to the compressor magnetically without rotary seals between the motor chamber and downhole environment. A multi-stage hybrid axial compressor capable of producing multiphase gas without sustaining performance inhibiting damage to the sleeve or blades of the compressor when operating with solids present due to the vertical flow path.

The downhole assembly demonstrates highly efficient and reliable performance by increasing suction pressure close to the perforations without pipe friction loss and eliminating failure-prone components of conventional artificial lift technologies, such as rotating seals, lubricating oil and motor pressure compensating chambers from this architecture. In this architecture, the system has proven 200% increased natural gas production and 70% increased recoverable reserves.

The enabling technologies demonstrated by the SCS are also available for use on the surface of gas wells through the Inline Compressor System (ICS). The ICS offers the significant advantages of producing gas and process liquid simultaneously from the wellhead through the compressor without the need for liquid

separation. It also improves safety compared to traditional surface wellhead compressors by eliminating the need for dynamic seals, therefore lowering H₂S exposure risk.

Introduction

Natural gas is both abundant and reliable, and infrastructure is already widely in place for transportation and storage, but an essential question remains – how can production and recoverability of natural gas be maximized while remaining competitive in the increasingly diverse energy landscape.

Previously, as a gas well's natural pressure depletes and gas is not able to efficiently flow to the wellbore, natural gas producers had only a couple options to continue producing. Rely on surface wellhead compressors (SWCs) or other artificial lift methods to boost production from aging wells whose natural pressure has diminished or spend significant upfront capital to explore new resources. While SWCs aid in extracting additional resources by reducing wellhead pressure, they require separation of liquids from the gas flow before compression.

It has been well recognized by industry that compression downhole, near gas well reservoirs, provides a significant increase in the rate of production relative to existing surface wellhead compressors but such systems must be able to withstand the downhole gas well environment. A high-speed, electric-driven downhole compressor (SCS) enables extended economic usefulness of gas wells.

The system simultaneously provides benefits of various artificial lift methods; such as positive or dynamic displacement pumps and gas lift, surfactants or demulsifiers; by injecting thermal and kinetic energy to the production stream to increase temperature changing the density and viscosity of produced fluids enabling gas and liquid lift. Downhole compression creates a suction effect at the compressor inlet, consequently lowering the bottom hole flowing pressure and increasing the fluid velocities. This drawdown actively induces more mass flow from the formation to the wellbore and facilitates the lowest possible abandonment pressure. When installed at the surface, the system provides the same benefits in terms of ability to handle multiphase flow and production increase by increasing wellhead velocity and in turn downhole velocity but has reduced installation time compared to a downhole installation and a significantly smaller footprint and improved safety compared to existing surface compressors.

Description of the System

This turbomachinery is comprised of a novel architecture that leverages robust, reliable and proven technologies to enhance gas well performance when installed in the downhole gas well environment or at the wellhead in place of SWCs. The Subsurface Compressor System™ (SCS) and Inline Compressor System™ (ICS) shown in [Figure 1](#) are comprised of three primary modules which ensure extended life and high-quality performance.

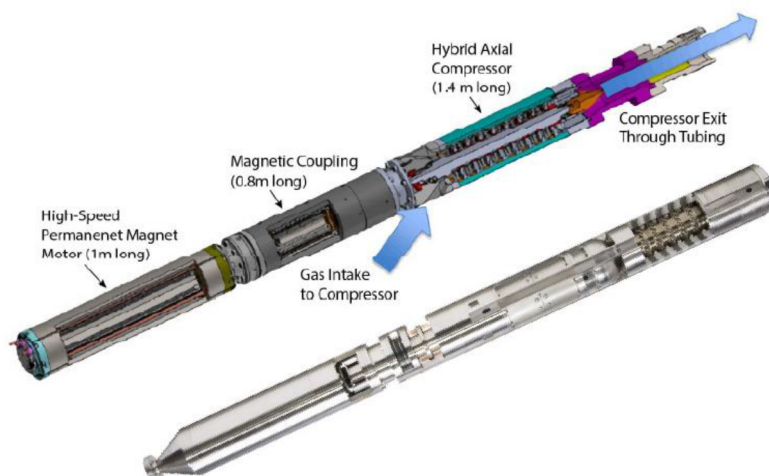


Figure 1—Compressor System

High-Speed Motor

The system is electrically driven by a hermetically sealed, power dense Permanent Magnet Motor (PMM) as shown in Figure 2, capable of operating at speeds over 50,000 RPM. The rotor section, which contains a series of magnets and other proprietary components, is sealed within Inconel 718. Copper wires for the stator are varnished and potted before the complete stator is inserted into a hermetically sealed chamber. Hermetically sealing the complete motor section of the system provides reliable operation by isolating all electrical components from gasses and liquids present in the external environment. The motor's high-speed operation is essential for compressible fluids as it facilitates efficient energy transfer and high throughput.

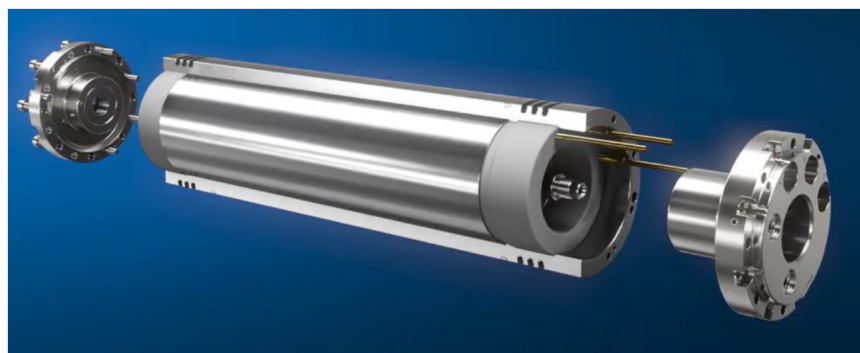


Figure 2—Motor Module of Compressor System

Magnetic Coupling

To maintain a hermetic seal of the motor, a large gap, high torque Magnetic Coupling was introduced as shown in Figure 3. Using of this coupling eliminates the protector from an artificial lift device while fully isolating the motor from fluids by an isolation can. A male-female coupling is placed on either side of the seal to effectively transmit torque out of the motor and to the system's compressor module. This creates a pressure seal which eliminates the possibility of process gas present in the well from getting into the motor. Mechanical couplings are not applicable in this case as they fit tightly around the shaft but cannot fully seal, thus leaving the possibility for gasses to enter the motor. A male magnetic coupling is situated at the end of the shaft to the compressor. Below the male coupling, there is an isolation can/barrier that hermetically seals everything below it from produced fluids. The space below the isolation can is filled with inert gas or air at roughly atmospheric pressure. A female magnetic coupling sits inside the inert gas and connects to the shaft of the motor. The torque and power from the motor is transmitted to the compressor from the female

magnetic coupling through the isolation can to the male magnetic coupling without the need of a physical connection between the shaft to the motor and the shaft to the compressor. The thrust and radial loads from the hydraulic section are supported by magnetic bearings, which completely levitate the shafts and prevent any physical contact between the rotating and stationary parts. In addition to supporting reliability of the system, the use of a magnetic coupling lowers cost and downtime associated with maintenance of mechanical couplings. Combining all the unique features of PMMs along with system level integration in the motors architecture maximizes availability and retrievability of downhole rotating equipment.



Figure 3—Coupling Module of Compressor System

Hybrid Wet Gas Compressor

The compressor module of the SCS and ICS shown in Figure 4 is composed of multiple stages of rotating and non-rotating hybridized axial compressor blades. The presence of particles (i.e. sands) and multiphase fluid (simultaneous flow of gasses and liquids) presents the potential for technologies operating in the environment to incur physical damage. The straight flow path and large gaps between the stator and rotor blades enable production of multiphase gas without sustaining physical damage to the sleeve or blades of the compressor when operating with solids up to 200 microns (μm) present. The axial flow allows material to travel vertically through the compressor as opposed to the perpendicular flow of a radial compressor, which would lead to significant abrasion from particles passing through the compressor. A combination of the compressors ability to increase reservoir drawdown, accelerate bottomhole gas velocity at the suction point and increase pressure and temperature of the gas at the discharge point enables significant liquid handling of up to 150 bbl/MMSCF and carryover to the surface, postponing liquid loading without the use of additional equipment such as pumps. Similar to electric submersible pump (ESP) systems for oil wells, compressor stages can be stacked to reach the desired pressure ratios at given flows and temperatures.

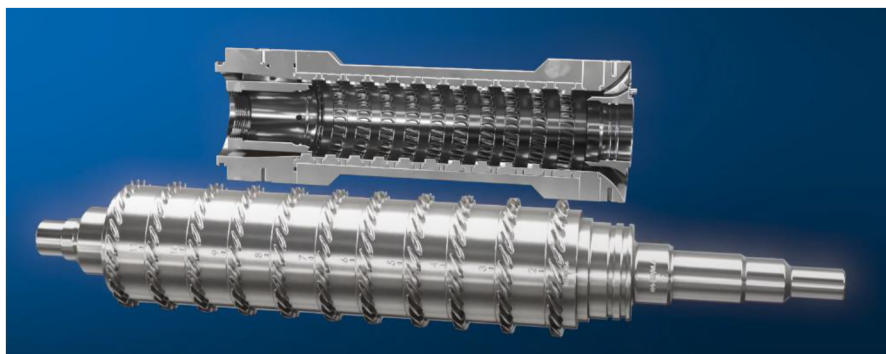


Figure 4—Compressor Module of Compressor System

The SCS and ICS were designed with the specific conditions and challenges of gas wells in mind. Together, its components allow for the system to operate reliably and continuously. The remainder of the system is similar to that of an ESP. On the topside resides the control center which houses the VSD, transformer, system level controller and monitoring equipment. Downhole installations are also similar to an ESP with conventional cables clamped outside the production tubing to transmit the power to the SCS downhole. A top level well completion with the SCS is shown in Figure 5. A top level well completion with the ICS is shown in Figure 6.

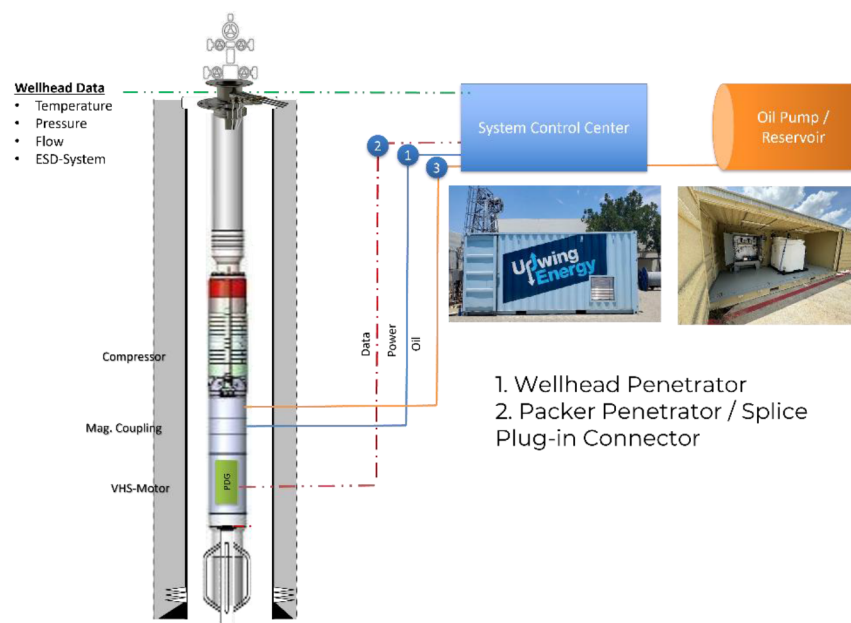


Figure 5—Completion of Subsurface Compressor System™ (SCS)

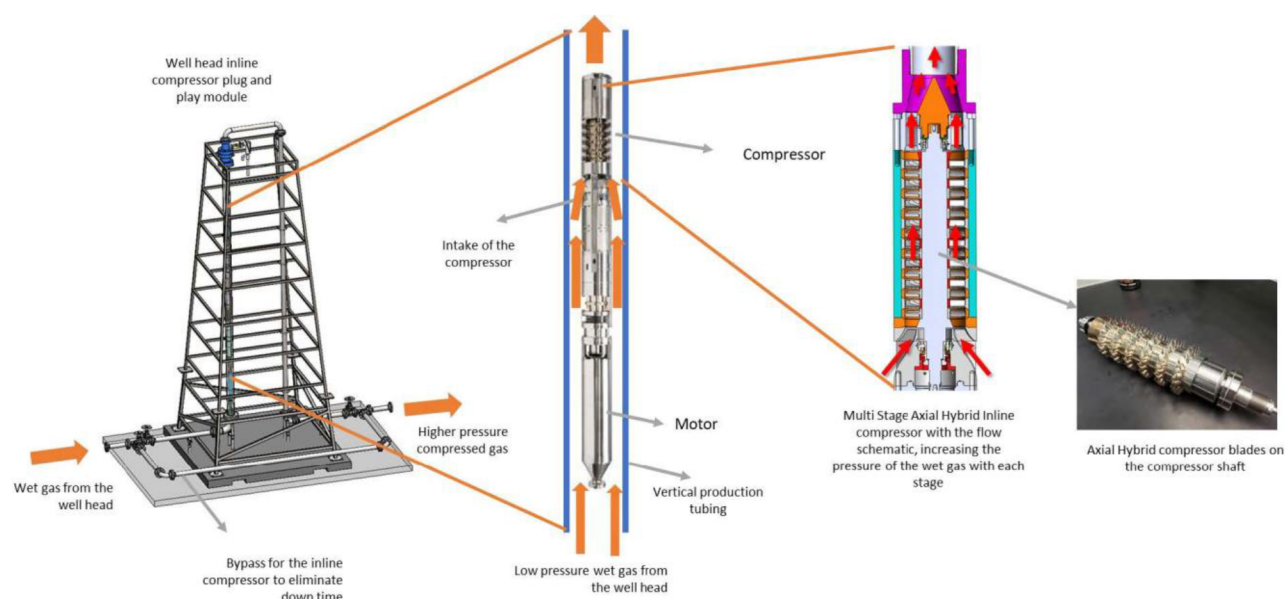


Figure 6—Completion of Inline Compressor System™ (ICS)

Benefits of the Compressor System in Pressure Depleted Reservoirs

In a newly producing gas well, a natural flow stream to the wellhead occurs as a result of the pressure differential between the reservoir and bottomhole. As the well matures, it gradually loses the momentum

from the reduction of the reservoir pressure because of pressure depletion, leading to a decreased driving force behind the gas. When a well reaches a predetermined abandonment pressure, it will become uneconomical to produce. However, quite often lack of momentum from fluid flows results in liquids dropping from the gas stream and accumulating in the wellbore causing an inability to naturally produce gas to the wellhead prior to the well reaching the predetermined abandonment pressure. At this stage in a gas wells life cycle, artificial lifts such as SWCs, pumping systems, plunger systems, high-pressure gas injection, etc. can be applied to delay ultimate abandonment.

It is well recognized by industry that compression downhole, near the gas well perforations, provides a significant increase in the rate of production compared to traditional SWCs. The SCS increases gas production by decreasing bottom hole flowing pressure and causing higher reservoir drawdown. This compression system provides suction effects at the compressor inlet, lowering intake pressure, and boosting effects at the compressor outlet, increasing discharge pressure as the gas flows to the wellhead. Due to the compressor being set near the bottom hole, where the gas is denser due to the higher downhole pressure compared to surface pressure, lower pressure is required at the compressor inlet to allow continues production even in depleted wells. In the case of SWCs due to the loss in pressure as the gas reaches the surface, high vacuum pressures are required at the compressor inlet, preventing effective operation. The same principle can be seen in the vehicle turbocharger, which provides compressed air with much higher density to boost air mass flow rate in the fixed cylinder volume, leading to the increase of engine power (Oh et al. 2018). In gas reservoirs, downhole compression will increase recoverable reserves by prolonging the production decline rate, postponing liquid loading, and providing lower abandonment pressure as shown in Figure 7.

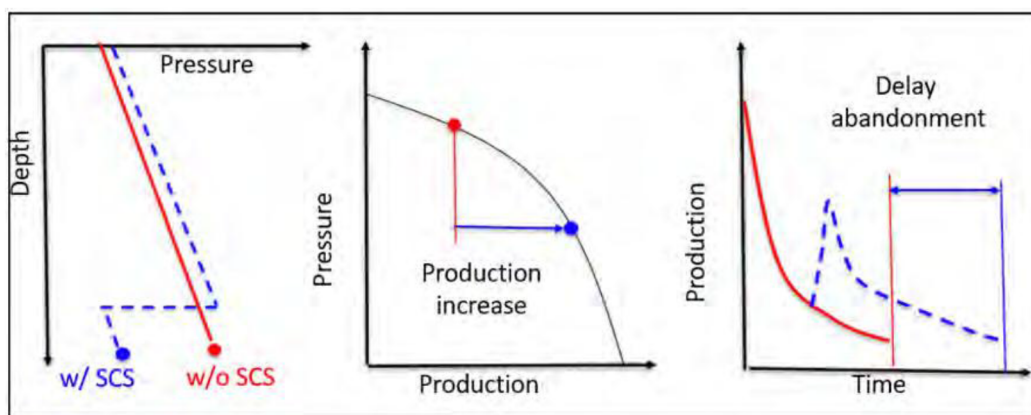


Figure 7—Effects of Downhole Compression

To further illustrate the benefits of subsurface compression on production and recoverable reserves compared to the same well producing naturally without artificial lift intervention, a simulation was carried out with conventional gas well data. Figure 8 presents the pressure profiles with and without downhole compression. The pressure profile of a gas well producing naturally is represented by the red line P_3 , gathering pressure, to P_1 , downhole flowing pressure. The pressure profile of a gas well with the downhole compressor system is represented by the blue line with gathering pressure (P_3) the same as producing without artificial lift and P_1 showing bottom hole flowing pressure which is significantly lower than in the case of natural production. The increased reservoir drawdown, represented by the difference between P_1 in red and P_1 in blue, by the subsurface compressor increases gas production flow from the formation into the wellbore according to the well responses to drawdowns, for example the Inflow Performance Relationship (IPR). The black dotted line represents pressure gradient required by a compressor on the surface to produce the same gas flow rate as with a subsurface compressor (Chen et al. 2019).

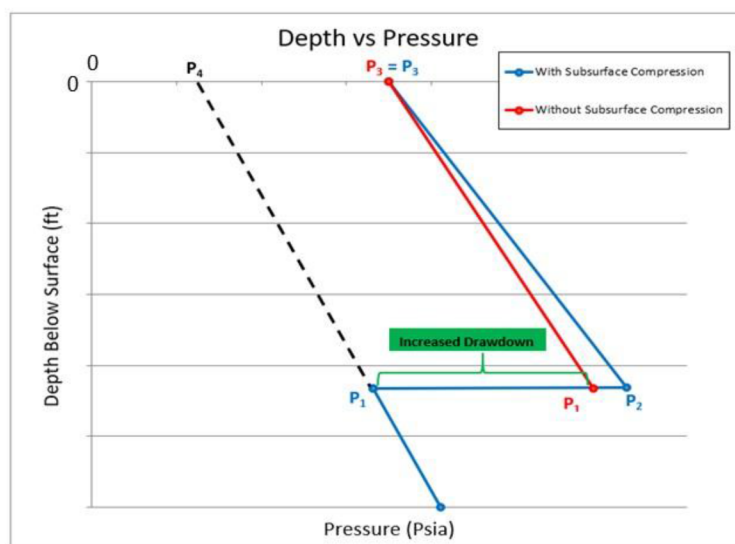


Figure 8—Pressure Profiles with and without Downhole Compression

Another advantage of the subsurface compressor is its capability to carry more liquids to the surface, abating the issue of liquid loading. This is accomplished by the system adding both external energy to the flow and changing the density of produced fluids. The increased gas velocity promotes carrying of liquids and reduced back pressure on the formation. The injected energy to the gas stream increases enthalpy at the compressor discharge enabling the evaporation of liquids present and delayed liquid loading of the formation.

The ICS provides the same benefits as the SCS by effectively increasing velocity of the gas and maintaining gathering line pressures through the compressors inlet suction and discharge boosting effects.

With the ICS being a fully integrated compression system, it addresses challenges traditional SWCs face by eliminating the ancillaries that are required for many SWCs. Consider the case of producing from a wet sour gas well. In this case, surface wellhead compression would require systems for dewatering and sweetening processes prior to the gas undergoing compression. This process has a large footprint due to the required infrastructure, presents safety risks through H₂S exposure and environmental risks by release of methane through compressor seals. An alternative option of operating an H₂S-rated SWC with soft material is available but this is highly costly and still requires dewatering of the flows. By being installed inline, the ICS acts as a wellbore extension thus intrinsically providing additional benefits not applicable in the case of SWCs. The enclosure of the compression system within the wellbore extension eliminates exposure to the atmosphere therefore improving personnel safety by eliminating risk of H₂S exposure. An H₂S sweetener is also not needed as the ICS uses the same metallurgy as the SCS, which is designed to withstand all conditions present in the downhole environment by being composed of high-strength, corrosion-resistant material. The ICS is able to produce the well without the need for liquid separation because of its axial compressor design which allows fluids to pass through the center of the compressor. Additionally, high-speed operation and temperature increase promote fluid atomization. The magnetic coupling used to transmit torque from the motor module to the compressor eliminates the need for a solid shaft between the two modules, thus providing the system with a seal-less design that eliminates the risk of methane leaks, and minimizes downtime and cost related to maintenance of seals present in SWCs. Figures 9 and 10 illustrate the key differences between a traditional surface wellhead compressor and the Inline Compressor System. Additional benefits include maintained pressure eliminating the need for flaring, smart controls and operation for safety, flexible inlet pressure operating range and efficient mixing of gas and liquid for delayed dropout to the downstream network.

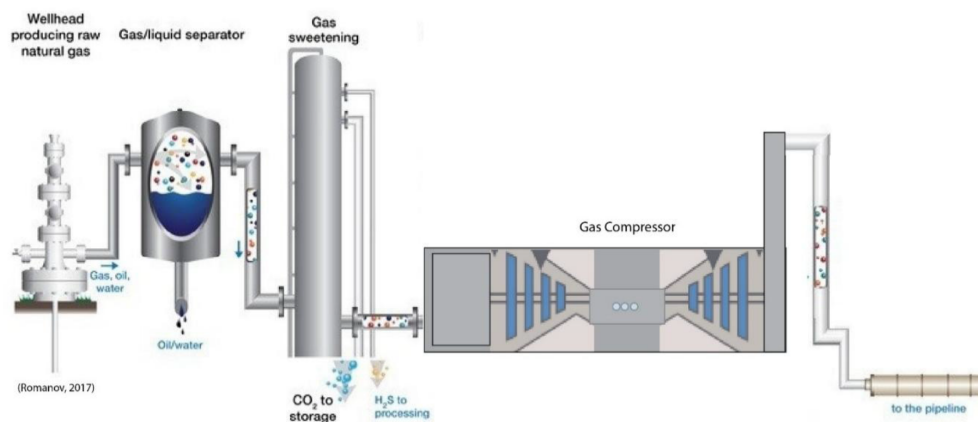


Figure 9—Traditional Surface Wellhead Compression of Sour Wet Gas

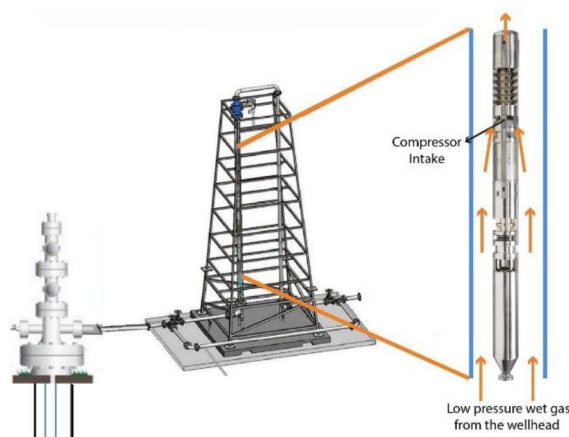


Figure 10—Inline Compression of Sour Wet Gas

Case Studies and Results

The novel axial compression technology described above and utilized in both downhole and surface applications has successfully undergone rigorous testing, field trials and evaluation to illustrate the impact of the system on gas production and recoverable reserves.

Case 1 – Field Testing

In order to assess the robustness and resilience of the compression technology and supporting Surface Control Center (SCC) equipment, the system undertook an extensive testing initiative in a conventional natural gas well. Emphasis was on subjecting the SCS, VSD in the SCC and the control algorithms to numerous start stop cycles.

The system was subjected to 1,024 autonomous start stop cycles to validate mechanical resilience and stability of the SCS while encountering various rigid body modes present during these cycles. This endurance test demonstrated outstanding system longevity and strong resistance to performance degradation. The testing also addressed the severe issue of electrical catch and spin by executing autonomous cycles while the well was flowing, demonstrating the electrical components' ability to withstand and recover from the intense start stop phases, ensuring operational longevity.

Testing provided a unique opportunity to evaluate the system level autonomous control scheme's ability to remotely operate the system during critical phases. In addition to providing monitoring and system adjustments as needed to ensure continuous operation, the system's autonomous electrical control

equipment, housed in the SCC, functioned seamlessly in each of the 1,024 startup and shutdown cycles conducted during the duration of this testing.

This testing further proved the equipment's ability to navigate through complex rigid body modes, withstand severe electrical challenges and function effectively without onsite personnel.

Case 2 – Onshore North Africa Intermittently Producing, Condensate Blocked

A downhole compression simulation study was conducted on a depleted, highly prolific onshore gas field in North Africa, targeting production enhancement from a sandstone reservoir nearing depletion. The reservoir, initially producing under depletion drive with no aquifer support, is now approaching a reservoir pressure of 1,000 psi and exhibits wellhead flowing pressures around 600 psi. Well performance decline curves, shown in Figure 11, illustrate the degradation caused by increased water production and clearly support this observation.

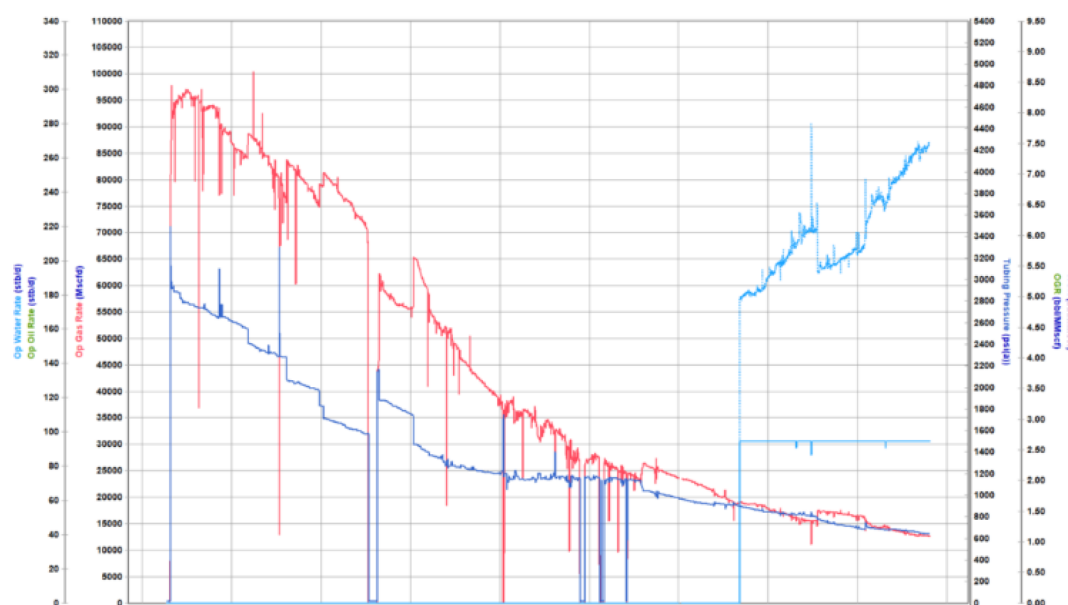


Figure 11—Well Performance Decline Curves

As production has continued, an increasing water-to-gas ratio has been observed, indicating rising mobile water saturation and impacting multiphase flow behavior within the wellbore. The evaluated wells, characterized by deviated trajectories, are experiencing deliquification issues that further degrade performance. Multiphase flow profiles highlighting liquid holdup zones in the deviated wellbores have been developed to better understand these challenges, as shown in Figure 12.

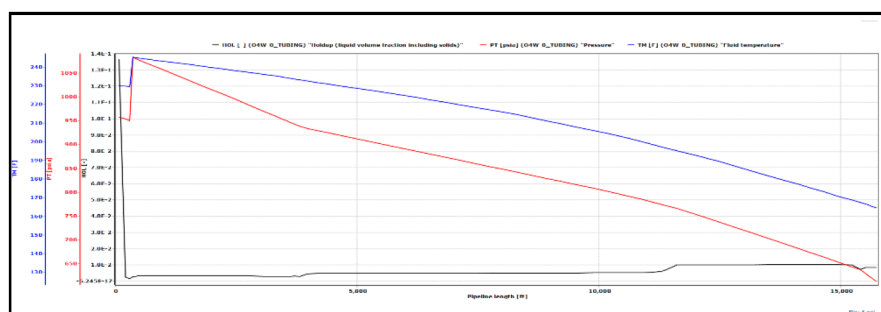


Figure 12—Multiphase Flow Profiles Highlighting Liquid Holdup Zones in the Deviated Wellbores

A comprehensive model was developed and calibrated using multirate separator well tests, rate transient analysis, and reservoir simulation. Pressure and rate transient analysis graphs, shown in Figure 13 below, were used to validate the accuracy of the model calibration. Multiphase transient flow modeling was employed to accurately capture downhole flow dynamics, identify liquid holdup zones, and quantify incremental gas recovery potential.

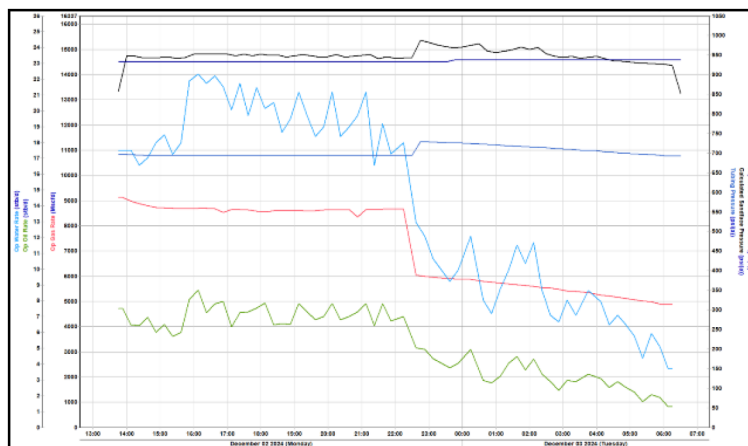


Figure 13—Pressure and Rate Transient Analysis Graph

Results indicate that the installation of surface compression can boost natural gas production by over 40%, delivering approximately 4.5 MMSCFD of additional gas per evaluated well. Beyond the immediate production gains, surface compression significantly lowers bottomhole pressure, enabling continued reservoir drawdown and extending the productive life of the wells in this maturing asset. Comparative production forecast graphs (Figure 14), with and without surface compression, clearly demonstrate the substantial incremental gas gains and extended reservoir life achievable through this intervention.

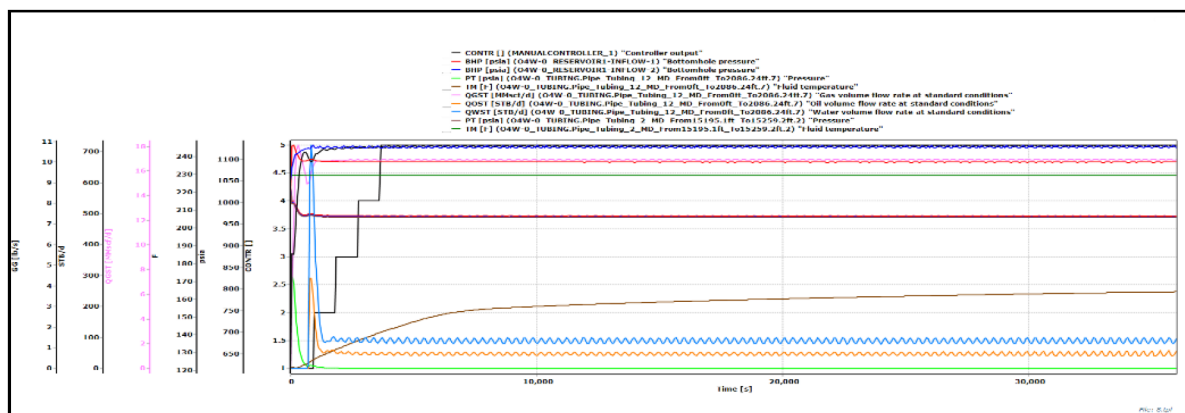


Figure 14—Comparative Production Forecasts with and without Downhole Compression

Case 4 – Conventional Onshore Pressure Depleted Gas Well Field Installation

This well was initially drilled in 1973 and operated by various companies until it ceased production in 2000. The field remained on the market for several years before being acquired by the current operator in 2018.

In 2021, a swabbing operation was conducted to evaluate the well's production potential. Despite several attempts, output remained limited to 30 MSCFD, as shown in the plot below (Figure 15). This constraint was largely due to significant reservoir depletion from extended production in surrounding wells, along with formation damage likely resulting from over 20 years of water saturation.

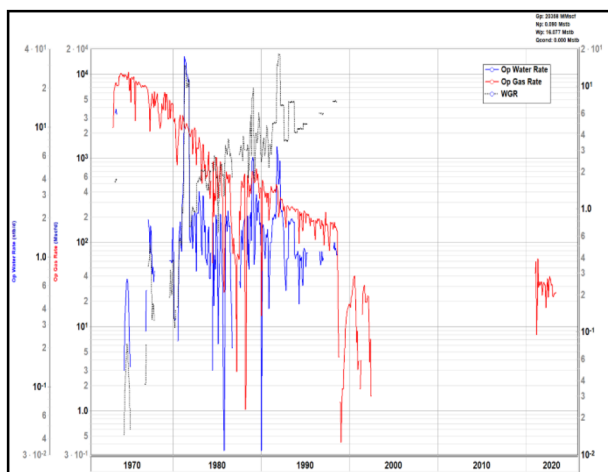


Figure 15—Original State Production Plot

In late 2023, the operator launched a workover operation using chemical treatments to mitigate the formation damage. The goal was to restore reservoir connectivity and improve productivity. Although the target was to reach 800 MSCFD, actual production fell short of expectations.

Testing was conducted to validate simulation results in comparison to actual SCS performance. A downhole completion was engineered to boost liquid and gas production while maintaining liquid unloading of the pressure-depleted reservoir. Positioned approximately 2,500 feet deep, the SCS features data collection gauges at various depths and centralizing equipment both above and below it to facilitate gas flow for cooling.

The deployment unfolds in three phases: 1) Receiving and pretest inspections, 2) SCS installation and run-in-hole (RIH), and 3) Surface commissioning and startup. Each phase adheres to industry standards. Upon receipt, the SCS undergoes electrical testing including a low-speed spin test and thorough vetting to ensure operational readiness. Measuring around 20 feet in length, the SCS is fully tested before being transported to the rig floor.

The SCS installation process parallels that of an ESP with a PMM, incorporating similar core components. Additional components, such as the standard XN Nipple and Plug, are used to prevent electrical energization during RIH. Key components include:

1. ESP Cable
2. TEC Cable
3. Wellheads and tubing hangers
4. Banding and clamping
5. Standard third-party equipment (e.g., desanders, packers)

The installation of the SCS delivered a notable improvement. The system enhanced water removal and allowed for minor condensate recovery, helping to optimize conditions for sustained production.

Results demonstrated close agreement between simulated and actual flow rates and pressures.

The SCS was designed to support higher potential flow rates, providing flexibility across a broad operating range and it proved instrumental in improving performance. Since its installation, the well has produced between 200 and 300 MSCFD, as illustrated in the plot below (Figure 16).

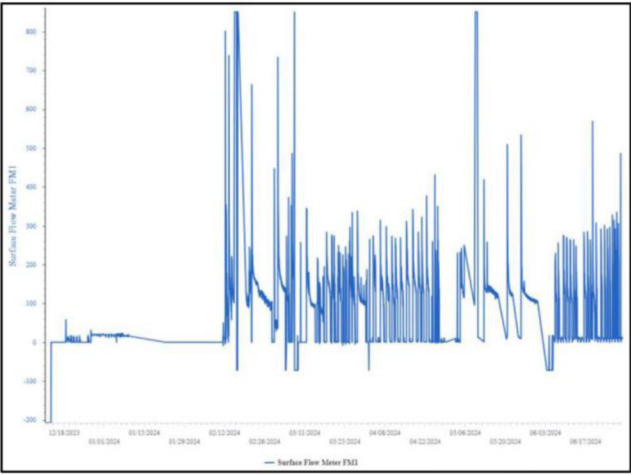


Figure 16—Gas Flow Rate with Downhole Compression

As the engineering team continues to collect and analyze tool data, the operations team leverages this information to gain deeper insight into reservoir and wellbore behavior. High-frequency downhole data, triggered by preset conditions, has proven valuable in understanding how the reservoir and wellbore respond to dynamic changes.

It is increasingly evident that the SCS not only enables varied operating conditions but also serves as a valuable reservoir management tool, supporting both pressure and rate transient analysis.

Conclusions

The benefits of the Subsurface Compressor System and Inline Compressor System can be summarized by Table 1 and Figure 17 below. The compression systems replace the vicious cycle of liquid loading with a virtuous cycle of liquid unloading by increasing gas velocity at both the intake and discharge which reduces back pressure and carries liquids present in the wellbore to the surface. The systems have demonstrated reliable performance and longevity in difficult to operate environments and have shown significant potential to extend the economic life of underperforming gas wells.

Table 1—Benefits of SCS and ICS Compared to Traditional Surface Wellhead Compressors

Decrease bottom hole pressure	→	Increase flow rate
Increase velocity	→	Liquid lifting, horizontal sweeping
Increase temperature	→	Evaporates volatiles
Increase availability	→	Minimizes pore clogging & deposition in wellbore
Increase topside pressure	→	Minimizes topside costs

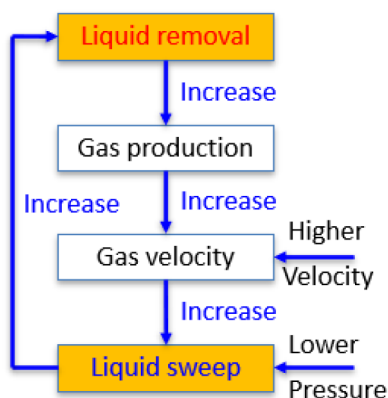


Figure 17—Benefits of SCS and ICS Compared to Traditional Artificial Lift Methods

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